

# Structures at six: the Eisenstein geometry of $E_6$ , a parity obstruction for bilinear $\text{AME}(4, d)$ constructions, and the anatomy of two $\text{AME}(4, 6)$ representatives

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## Abstract

Local dimension six carries two famous exceptions: there is no pair of orthogonal Latin squares of order six (Tarry), and the maximal number of mutually unbiased bases in  $\mathbb{C}^6$  remains unknown. The first exception was resolved quantum-mechanically in 2022 by the construction of an absolutely maximally entangled state  $\text{AME}(4, 6)$ , equivalently a 2-unitary matrix of order 36. This note collects four independent results, all verified computationally from first principles.

(i) We give an explicit realization of the  $\mathbb{Z}[\omega]$ -module structure of the  $E_6$  root lattice through the fourth power of a Coxeter element, and verify that the resulting 12 complex lines in  $\mathbb{C}^3$  decompose into exactly 4 orthogonal frames that are pairwise mutually unbiased. Thus  $E_6$  realizes the *complete* MUB system of a qutrit, and the count of positive roots factors as  $36 = 12 \times 3$ .

(ii) We prove a parity obstruction: a flat 2-unitary of order  $d^2$  whose phase is a bilinear form over  $\mathbb{Z}/d$  exists *if and only if*  $d$  is odd. The proof is two lines and uses only that the units of  $\mathbb{Z}/d$  are odd for even  $d$ . This explains why the finite-field construction available at every odd  $d$  has no analogue at  $d = 6$ , and shows the obstruction is one of parity rather than of non-prime-power order.

(iii) We reconstruct and independently verify two Butson-type  $\text{AME}(4, 6)$  representatives with all phases in  $\mu_6 = \mathbb{Z}[\omega]^\times$ , and show that their phase function is not a polynomial of degree  $\leq 2$  over  $\mathbb{Z}/6$ , is not translation covariant, and does not factor through the Chinese Remainder decomposition  $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$ .

(iv) We compute the automorphism group of the golden  $\text{AME}(4, 6)$  representative and find it to be  $\mathbb{Z}/2$  at the level of the modulus pattern, with the nontrivial element *failing* to lift to a monomial symmetry of the state. In particular the natural conjecture that the golden state is covariant for the exceptional action of  $\text{PGL}_2(\mathbb{F}_5)$  on the six points of  $\mathbb{P}^1(\mathbb{F}_5)$  is false for this representative.

## 1 Introduction

Two exceptional phenomena meet at local dimension six.

The first is combinatorial. Euler conjectured in 1779, and Tarry proved in 1900, that no two orthogonal Latin squares of order six exist; 6 is the unique order  $> 2$  for which this fails [1, 2, 3]. The best classical configuration attains 34 of the 36 officer pairs. In 2022 Rather et al. [4] showed that the problem admits a quantum solution: an absolutely maximally entangled state of four six-level systems,  $\text{AME}(4, 6)$ , equivalently a 2-unitary matrix of order 36, equivalently a perfect tensor with four legs of dimension six. The amplitudes of their solution involve the

golden ratio  $\varphi = (1 + \sqrt{5})/2$  and phases that are 20th roots of unity; it is known as the *golden* AME state. Subsequently it was shown that infinitely many inequivalent solutions exist [6], and two further constructions produced representatives that are Butson-type complex Hadamard matrices with all 1296 entries of equal modulus and phases in  $\mu_6$  [7, 8].

The second exception is measurement-theoretic. In prime-power dimension  $d$  there exist  $d + 1$  mutually unbiased bases (MUBs), and this is maximal. In dimension six only three MUBs are known, and whether four exist is a long-standing open problem. Six is the smallest dimension where the question is open.

That the same dimension hosts both exceptions is folklore-level motivation for asking whether the objects that repair the first say anything about the second. This note is a report of what we could establish by direct computation, kept scrupulously separate from what we could not. Section 3 treats an Eisenstein structure in dimension three that turns out to be exactly a complete MUB system; Section 4 isolates the arithmetic obstruction at six and shows it is a parity phenomenon valid for all even  $d$ ; Section 5 dissects two AME(4, 6) representatives. Appendix A tabulates the epistemic status of every claim.

All computations were performed independently of the cited literature: root systems were generated from Cartan matrices by breadth-first reflection closure, and the AME matrices were rebuilt from their defining data and re-verified, rather than assumed.

## 2 Preliminaries

Let  $d \geq 2$  and write  $N = d^2$ . Index  $\mathbb{C}^N \otimes \mathbb{C}^N$  by four indices, and for  $X \in \mathbb{C}^{N \times N}$  define the *reshuffling* and *partial transpose*

$$X_{jk;lm}^R = X_{jl;km}, \quad X_{jk;lm}^\Gamma = X_{jm;lk}.$$

**Definition 1.**  $U \in \mathbb{U}(d^2)$  is *2-unitary* if  $U$ ,  $U^R$  and  $U^\Gamma$  are all unitary. Equivalently the associated four-index tensor  $\psi_{jklm} = d^{-1}U_{jk;lm}$  is a *perfect tensor*: every reduction to two of the four legs is maximally mixed,  $\text{tr } \rho^2 = d^{-2}$  for all six bipartitions. Equivalently  $|\psi\rangle \in \text{AME}(4, d)$  [10].

**Definition 2.**  $H$  of order  $N$  is a *complex Hadamard matrix* if  $HH^\dagger = N\mathbb{I}$  and  $|H_{jk}| = 1$  for all  $j, k$ ; it is of *Butson type*  $\text{BH}(N, q)$  if in addition every entry is a  $q$ -th root of unity [12].

We call a 2-unitary *flat* if all  $d^2 \times d^2$  entries have modulus  $d^{-1}$ , i.e. if  $dU$  is a complex Hadamard matrix. It is known that 2-unitaries do not exist for  $d = 2$  [11]; for  $d \neq 2, 6$  they are furnished by orthogonal Latin squares as permutation matrices, and  $d = 6$  is the case resolved in [4].

Throughout,  $\omega = e^{2\pi i/3}$ ,  $\mathbb{Z}[\omega]$  is the ring of Eisenstein integers,  $\mu_k$  is the group of  $k$ -th roots of unity, and  $\mathbb{Z}[\omega]^\times = \mu_6$ .

## 3 The Eisenstein geometry of $E_6$ and the qutrit MUB system

The number 36 occurs twice in our story: as  $|\Phi^+(E_6)|$  and as  $6^2$ . The following makes the first occurrence structurally transparent. That the  $E_6$  root system carries a  $\mathbb{Z}[\omega]$ -structure related to the Hessian configuration of 12 lines is classical, arising from the theory of complex reflection groups; that the Hesse configuration encodes the qutrit MUB system is known in quantum information. Our contribution is an explicit, uniform realization through the Coxeter element together with complete verification of the resulting geometry.

**Theorem 3.** *Let  $\Phi$  be the root system of  $E_6$  with  $W = W(E_6)$ , Coxeter element  $c$ , and invariant bilinear form  $B$  normalized so that  $B(r, r) = 2$  for roots. Put  $g = c^4$ . Then:*

- (a)  $g^3 = 1$ ,  $g$  preserves  $B$ , and  $g$  is fixed-point-free:  $\det(g - 1) = 27 \neq 0$ .
- (b)  $J := (2g + 1)/\sqrt{3}$  satisfies  $J^2 = -1$  and  $B(Ju, v) = -B(u, Jv)$ , so  $h(u, v) := B(u, v) + iB(u, Jv)$  is a Hermitian form making the ambient space  $\mathbb{C}^3$  and the  $E_6$  lattice a rank-3  $\mathbb{Z}[\omega]$ -module, with  $h(r, r) = 2$  for roots.
- (c) The group  $\mu_6 = \{\pm 1, \pm g, \pm g^2\}$  acts freely on the 72 roots, with exactly 12 orbits of size 6. Hence

$$72 = 12 \times 6 = \#\{\omega\text{-lines}\} \times |\mathbb{Z}[\omega]^\times|, \quad 36 = |\Phi^+| = 12 \times 3.$$

- (d) Let  $L_1, \dots, L_{12} \subset \mathbb{C}^3$  be the resulting complex lines. Then for  $i \neq j$  the quantity  $|h(r_i, r_j)|^2$  takes exactly two values, 0 and  $4/3$ . Each line is orthogonal to exactly two others, so the 12 lines partition into 4 orthogonal frames of 3; and for lines in different frames the normalized overlap is

$$\frac{|h(r_i, r_j)|^2}{h(r_i, r_i)h(r_j, r_j)} = \frac{4/3}{4} = \frac{1}{3}.$$

Consequently the 12 lines form  $d + 1 = 4$  pairwise mutually unbiased bases of  $\mathbb{C}^3$ : the complete MUB system of a qutrit.

*Proof (computational).* The 72 roots were generated from the  $E_6$  Cartan matrix by closing the six simple roots under simple reflections; all have  $B$ -norm 2. The Coxeter element has order 12, consistent with  $h(E_6) = 12$ ; since the exponents of  $E_6$  are 1, 4, 5, 7, 8, 11, none is divisible by 3, so  $c^4$  has no eigenvalue 1 and is fixed-point-free, which is what (a) records numerically as  $\det(g - 1) = 27$ . Items (b)–(d) were verified entrywise; the orthogonality graph on the 12 lines is 3-regular with components  $\{0, 7, 11\}, \{1, 6, 10\}, \{2, 5, 8\}, \{3, 4, 9\}$  in the ordering produced by our enumeration, and all  $\binom{12}{2} - 4\binom{3}{2} = 66 - 12 = 54$  cross-frame pairs give exactly  $1/3$ .  $\square$

*Remark 4.* The identity  $g^2 + g + 1 = 0$  on the ambient space is what allows  $J = (2g + 1)/\sqrt{3}$  to square to  $-1$ ; the factor  $\sqrt{3}$  is the different of  $\mathbb{Q}(\omega)$ , and  $\det(g - 1) = 27 = 3^3$  is the norm of  $(1 - \omega)^3$  up to units. The chain  $A_1 (h=2)$ ,  $A_2 (h=3)$ ,  $G_2 (h=6)$ ,  $F_4 (h=12)$ ,  $E_8 (h=30)$  of root systems whose exponents are exactly the totatives of  $h$  was also verified; it places the mod-6 unit pattern  $\{1, 5\}$  as the exponent set of  $G_2$ .

## 4 A parity obstruction for bilinear 2-unitaries

At every odd  $d$  there is a one-line construction of a perfect tensor,  $|\psi\rangle \propto \sum_{i,j} |i, j, i+j, i+2j\rangle$  over  $\mathbb{Z}/d$ . Its Fourier dual is a flat 2-unitary whose phase is a bilinear form. The following theorem determines exactly when such an object exists, and shows the failure at  $d = 6$  is a parity phenomenon.

For  $M = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \in M_2(\mathbb{Z}/d)$  and  $\zeta = e^{2\pi i/d}$ , define  $U^{(M)} \in \mathbb{C}^{d^2 \times d^2}$  by

$$U_{(x,y),(u,v)}^{(M)} = \frac{1}{d} \zeta^{x(m_{11}u + m_{12}v) + y(m_{21}u + m_{22}v)}. \quad (1)$$

**Lemma 5.**  $U^{(M)}$  is 2-unitary if and only if all three of

$$\det M = m_{11}m_{22} - m_{12}m_{21}, \quad m_{11}m_{22}, \quad m_{12}m_{21}$$

are units of  $\mathbb{Z}/d$ .

| $d$ | parity | $\#\{M \text{ admissible}\}$ | canonical solution                      |
|-----|--------|------------------------------|-----------------------------------------|
| 2   | even   | 0                            | —                                       |
| 3   | odd    | 8                            | $(1, 1; 1, 2)$ verified 2-unitary, flat |
| 4   | even   | 0                            | —                                       |
| 5   | odd    | 192                          | $(1, 1; 1, 2)$ verified                 |
| 6   | even   | 0                            | —                                       |
| 7   | odd    | 1080                         | $(1, 1; 1, 2)$ verified                 |
| 8   | even   | 0                            | —                                       |
| 9   | odd    | 648                          | $(1, 1; 1, 2)$ verified                 |
| 10  | even   | 0                            | —                                       |
| 11  | odd    | 9000                         | $(1, 1; 1, 2)$ verified                 |
| 12  | even   | 0                            | —                                       |

Table 1: Exhaustive census of bilinear phase forms  $M \in M_2(\mathbb{Z}/d)$  yielding a flat 2-unitary of order  $d^2$ . Existence holds exactly for odd  $d$ , in agreement with Theorem 6. Admissibility was tested by the criterion of Lemma 5 and confirmed numerically on representatives.

*Proof.* Write the exponent as  $E = m_{11}xu + m_{12}xv + m_{21}yu + m_{22}yv$ .

*Unitarity of  $U$ .* For row indices  $(x, y), (x', y')$  with  $(s, t) = (x - x', y - y')$ ,

$$(UU^\dagger)_{(x,y),(x',y')} = \frac{1}{d^2} \sum_{u,v} \zeta^{(sm_{11}+tm_{21})u} \zeta^{(sm_{12}+tm_{22})v},$$

which by orthogonality of characters vanishes unless  $(s, t)M \equiv 0$ , in which case it equals 1. Hence  $U$  is unitary iff  $(s, t)M \equiv 0 \Rightarrow (s, t) = 0$ , i.e. iff  $\det M \in (\mathbb{Z}/d)^\times$ .

*Unitarity of  $U^R$ .* Reading  $E$  with row index  $(x, u)$  and column index  $(y, v)$ , the terms  $m_{11}xu$  and  $m_{22}yv$  depend only on the row, respectively only on the column; they contribute diagonal unitary factors on either side and do not affect unitarity. The genuine pairing between row and column is  $m_{12}xv + m_{21}yu$ , of the form (1) with matrix  $\begin{pmatrix} 0 & m_{12} \\ m_{21} & 0 \end{pmatrix}$ , whose determinant is  $-m_{12}m_{21}$ . So  $U^R$  is unitary iff  $m_{12}m_{21}$  is a unit.

*Unitarity of  $U^\Gamma$ .* Reading  $E$  with row index  $(x, v)$  and column index  $(u, y)$ , the terms  $m_{12}xv$  and  $m_{21}yu$  are row-only and column-only respectively, and the pairing matrix is  $\text{diag}(m_{11}, m_{22})$  with determinant  $m_{11}m_{22}$ . So  $U^\Gamma$  is unitary iff  $m_{11}m_{22}$  is a unit.  $\square$

**Theorem 6** (Parity obstruction). *A matrix  $M \in M_2(\mathbb{Z}/d)$  with  $U^{(M)}$  2-unitary exists if and only if  $d$  is odd. For odd  $d$  one may take  $M = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ .*

*Proof.* Suppose  $d$  is even. Every unit of  $\mathbb{Z}/d$  is odd. By Lemma 5 both  $m_{11}m_{22}$  and  $m_{12}m_{21}$  are units, hence odd; therefore  $\det M = m_{11}m_{22} - m_{12}m_{21}$  is a difference of two odd numbers, hence even, hence not a unit of  $\mathbb{Z}/d$ . This contradicts Lemma 5.

If  $d$  is odd, take  $M = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ : then  $m_{11}m_{22} = 2$ ,  $m_{12}m_{21} = 1$  and  $\det M = 1$ , and 2 is a unit modulo any odd  $d$ .  $\square$

We verified Theorem 6 by exhaustive search over all  $d^4$  matrices for  $2 \leq d \leq 12$ ; Table 1 records the census. Note that the obstruction is insensitive to primality: it holds at  $d = 4, 8, 10, 12$  as well as at  $d = 6$ , and *fails* to obstruct at the odd composites  $d = 9, 15$ , where flat bilinear 2-unitaries do exist.

**Corollary 7.** *Any flat 2-unitary of order 36 — in particular any Butson-type AME(4, 6) representative — has a phase function that is not a bilinear form over  $\mathbb{Z}/6$ .*

This gives a precise sense in which the  $d = 6$  solutions must be “non-Gaussian”: the quadratic-Gauss-sum mechanism that manufactures perfect tensors at every odd  $d$  is unavailable, and something genuinely different (in the known constructions, biunimodular vectors [8, 9]) must take its place.

## 5 Anatomy of two AME(4, 6) representatives

### 5.1 The golden representative

We parsed the analytic form of the golden AME(4, 6) matrix from the authors’ own source code and re-verified it independently. It has 112 nonzero entries with three amplitude values

$$a = \frac{1}{\sqrt{5+\sqrt{5}}}, \quad b = \frac{\sqrt{5+\sqrt{5}}}{\sqrt{20}}, \quad c = \frac{1}{\sqrt{2}},$$

and phases in  $\mu_{20}$ ,  $w = e^{i\pi/10}$ . We confirm  $b/a = \varphi$  exactly and  $a^2 + b^2 = c^2 = \frac{1}{2}$ ; that  $U$ ,  $U^R$ ,  $U^\Gamma$  are unitary; and that all six bipartition purities equal  $1/36$  to machine precision. The amplitude census is 40 entries of modulus  $a$ , 40 of modulus  $b$ , and 32 of modulus  $c$ .

Motivated by the exceptional 2-transitive action of  $\mathrm{PGL}_2(\mathbb{F}_5) \supset \mathrm{PSL}_2(\mathbb{F}_5) \cong A_5$  on the six points of  $\mathbb{P}^1(\mathbb{F}_5)$  — the source of the outer automorphism of  $S_6$ , and of golden character values — it is natural to ask whether the golden state is covariant for a copy of  $A_5$  acting on the six levels. We tested this exhaustively.

**Proposition 8.** *Let  $\mathcal{E} \subset (\mathbb{Z}/6)^4$  be the support of the golden representative, colored by amplitude class. The group of quadruples  $(\pi_1, \pi_2, \pi_3, \pi_4) \in S_6^4$  acting independently on the four legs and preserving  $\mathcal{E}$  together with its coloring has order 2. Its nontrivial element is*

$$\pi_1 = (0\,1)(2\,3)(4\,5), \quad \pi_2 = (0\,5)(1\,4)(2\,3), \quad \pi_3 = \pi_4 = \mathrm{id}.$$

Moreover this element does not lift to a monomial symmetry: there are no diagonal unitaries  $D_L, D_R$  with  $D_L(P_{\pi_1} \otimes P_{\pi_2}) U D_R = U$ .

*Proof (computational).* Leg-wise color profiles reduce the search to 48, 48, 4, 8 candidates on the four legs; pair-profile filtering and constraint propagation on the fibers leave 2 quadruples, both of which are then verified to preserve all 112 colored edges. For the second claim, write  $V = (P_{\pi_1} \otimes P_{\pi_2})U$ ; then  $|V| = |U|$ , and a monomial lift exists iff the phase ratio  $\arg(V_{jk}/U_{jk})$  is of the separable form  $r_j + c_k$  on the support. The support bipartite graph has 9 connected components, each of size 8; performing the propagation *within each component* (the assignment is free between components) yields residuals up to 2.51 radians, so no separable solution exists.  $\square$

**Corollary 9.** *The golden representative is not covariant for any nontrivial group of level permutations acting monomially; in particular it carries no  $A_5$  or  $\mathrm{PGL}_2(\mathbb{F}_5)$  symmetry.*

*Remark 10 (Scope).* This is a statement about *this representative*. AME states are defined up to local unitaries, and a symmetry invisible in one frame may exist in another; the negative result rules out permutation-monomial symmetry in the analytic frame of [4], not in the whole LU class. We flag the possibility that the golden solution’s arithmetic  $(\mu_{20}, \varphi)$  is a feature of the seed — the algorithm of [4] is initialized near the best classical permutation [14] — rather than of the state.

## 5.2 The Butson (Eisenstein) representatives

We rebuilt the biunimodular construction of [7, 8] from scratch: with  $F_6$  the Fourier matrix,  $P$  the mod-6 generalized CNOT, and  $K = (F_6 \otimes \mathbb{I})P(F_6 \otimes \mathbb{I})$ ,  $L = (F_6 \otimes \mathbb{I})P(F_6^\dagger \otimes \mathbb{I})$ , we form  $\mathcal{U}_j = K \text{diag}(e^{2\pi i \Lambda_j}) L$ . We verify  $K = K^\dagger$ ,  $L = L^T$ , and that both are flat.

**Proposition 11.**  $\mathcal{U}_1$  and  $\mathcal{U}_2$  are flat 2-unitary matrices of order 36: all entries have modulus  $1/6$ , all of  $\mathcal{U}$ ,  $\mathcal{U}^R$ ,  $\mathcal{U}^T$  are unitary, and all six bipartition purities equal  $1/36$ . Every entry is exactly a sixth root of unity: writing  $\mathcal{U}_{jk} = \frac{1}{6}\zeta_6^{m_{jk}}$  with  $m_{jk} \in \mathbb{Z}/6$ , the reconstruction error is  $8.3 \times 10^{-15}$ . Hence  $6\mathcal{U}_j \in \text{BH}(36, 6)$  and all phases lie in  $\mathbb{Z}[\omega]^\times$ .

For  $\mathcal{U}_1$  the exponent census over  $\mathbb{Z}/6$  is  $(204, 180, 294, 180, 204, 234)$  for  $m = 0, \dots, 5$ . We then probed the structure of the phase function  $m(x, y, u, v)$  on  $(\mathbb{Z}/6)^4$ .

*Observation 12.* For  $\mathcal{U}_1$ :

- (a) *Non-polynomiality.*  $m$  is not a polynomial of total degree  $\leq 2$  over  $\mathbb{Z}/6$ : fitting the 15 monomials of degree  $\leq 2$  in four variables against all 1296 points is inconsistent both modulo 2 (rank 11) and modulo 3 (rank 15).
- (b) *No translation covariance.* None of the shifts  $(x, u) \mapsto (x+t, u+t)$ ,  $(y, v) \mapsto (y+t, v+t)$ , or a simultaneous shift of all four legs changes  $m$  by a constant, for any  $t = 1, \dots, 5$ .
- (c) *Distance from bilinearity.* The best pure bilinear form agrees with  $m$  on 372 of 1296 points (28.7%, against a 16.7% random baseline), attained at  $M = (2, 3; 1, 2)$ .
- (d) *CRT non-factorization.* Write  $\zeta_6^m = (-1)^m \bar{\omega}^m$ , splitting the phase matrix into a  $\chi_2$ -layer  $S \in \{\pm 1\}^{36 \times 36}$  and a  $\chi_3$ -layer  $W \in \mu_3^{36 \times 36}$ . Neither layer is unitary after normalization: the Gram spectrum of  $S/6$  runs from 0 (so  $S$  is singular) to 9, and that of  $W/6$  from  $1/3$  to 7.

Part (d) is the structurally sharpest: the solution cannot be assembled from a real Hadamard factor and an Eisenstein factor separately. Together with Theorem 6 it says that the 2-adic and 3-adic layers of  $\mathbb{Z}/6$  are irreducibly entangled in any flat order-36 solution — the same factorization  $6 = 2 \cdot 3$  that destroys the field  $\mathbb{F}_6$  and hence the bilinear route also forbids the solution from splitting along that factorization. Parts (a)–(c) confirm that the phase function is genuinely of “biunimodular” rather than Gauss-sum type.

*Remark 13.* Our transcription of the third vector  $\Lambda_3$  from the published PDF failed its own biunimodularity test; we attribute this to text-extraction corruption on our side rather than to an error in [7], and have therefore excluded  $\mathcal{U}_3$  from Proposition 11. The authors’ machine-readable data is available at their repository.

## 6 Discussion and open problems

**A hexit holographic brick.** A verified four-leg perfect tensor of local dimension six is exactly the ingredient required to build a HaPPY-type holographic code [13] on six-dimensional legs. Unlike the qubit pentagon code, such a brick cannot arise from a stabilizer/OLS construction, by Theorem 6 together with the nonexistence of OLS(6). Constructing the associated  $((3, 6, 2))_6$  code explicitly and studying entanglement wedges on hexit legs is the natural next computation.

**The MUB(6) wall.** Theorem 3 exhibits, in dimension 3, an exact coincidence between an Eisenstein lattice structure and a complete MUB system. The question that this note raises but does not answer is whether the Butson representatives of Proposition 11 contain unbiasedness data in dimension six. A concrete program: decompose  $6\mathcal{U}_1 \in \text{BH}(36, 6)$  into its 36 blocks of

size  $6 \times 6$  and test which pairs of blocks are unbiased. We regard Observation 12(d) as mild evidence *against* a clean outcome: an object that refuses to factor along  $6 = 2 \cdot 3$  is unlikely to hand over a  $\mathbb{Z}/6$ -structured set of bases. We state this as an open problem rather than a conjecture.

**On the golden ratio.** The existence of the Butson solutions shows that  $\varphi$  is not forced by  $\text{AME}(4, 6)$ : there are representatives whose entire coefficient field is  $\mathbb{Q}(\zeta_6) = \mathbb{Q}(\sqrt{-3})$ , arithmetically disjoint from the golden field  $\mathbb{Q}(\zeta_{20})$  since  $\mathbb{Q}(\zeta_{20}) \cap \mathbb{Q}(\zeta_6) = \mathbb{Q}$ . Combined with Proposition 8, which removes the most natural mechanism by which  $\varphi$  might have entered (icosahedral character values via  $\mathbb{P}^1(\mathbb{F}_5)$ ), we conclude that the golden ratio in [4] is best regarded as an artifact of the seeded numerical continuation rather than as an invariant of the  $\text{AME}(4, 6)$  moduli space. A sharper formulation of the remaining question: does there exist a functional on the space of  $\text{AME}(4, 6)$  states, such as distance to the classical boundary, whose critical points are the golden and Butson representatives?

## A Status of claims

| Claim                                                                           | Status                                                   | Evidence                                                                       |
|---------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------|
| Lemma 5, Theorem 6 (parity obstruction)                                         | <b>proved</b>                                            | character-sum argument; exhaustive check $2 \leq d \leq 12$                    |
| Theorem 3 (a)–(d), $E_6$ MUB decomposition                                      | verified computationally; underlying structure classical | explicit construction from Cartan matrix; full $12 \times 12$ overlap table    |
| Proposition 11 ( $\mathcal{U}_1, \mathcal{U}_2$ are Butson $\text{AME}(4, 6)$ ) | verified (reproduces [7, 8])                             | rebuilt from $F_6$ and CNOT; purities and exact $\mu_6$ entries                |
| Proposition 8 ( $\text{Aut} = \mathbb{Z}/2$ , no monomial lift)                 | verified for this representative                         | exhaustive leg search; component-wise separability test                        |
| Observation 12 (a)–(d)                                                          | verified numerically                                     | rank/consistency over $\mathbb{F}_2, \mathbb{F}_3$ ; shift tests; Gram spectra |
| Golden $\varphi$ is an artifact of the seed                                     | <b>conjectural</b>                                       | supported by Prop. 8 and existence of Butson solutions; not proved             |
| MUB(6) data inside $\mathcal{U}_1$                                              | <b>open</b>                                              | not tested here                                                                |

## B Reproducibility

All results were obtained with NumPy in exact-integer or double precision as appropriate. Root systems were generated by reflection closure from the Cartan matrix; AME matrices were rebuilt from published defining data ( $\Lambda_j$  vectors, or the analytic entry list) and re-verified rather than trusted. Tolerances:  $10^{-9}$  for unitarity and purity assertions; observed errors were at the  $10^{-15}$  level. Exhaustive searches (Table 1, Proposition 8) enumerate their full stated domains.

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